

MATH 1210 Worksheet 1 Summer 2023

Suggested exercises, more exercises can be found in the textbook or on Dr. R.S.D. Thomas' webpage for archived material:

<http://home.cc.umanitoba.ca/~thomas/Courses>

Questions marked with * are more difficult.

1. Use induction to show that for $n \geq 1$ we have

$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

Solution: Let us denote by $P(n)$ the statement

$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

We want to show that $P(n)$ is valid for any $n \geq 1$.

Base case for $n = 1$: the left-hand side is $1^2 = 1$. The right-hand side is $\frac{1 \cdot (1+1) \cdot (2+1)}{6} = 1$. Hence the equation is valid for $n = 1$.

Induction step: We suppose that for some integer $k \geq 1$, the statement $P(k)$ is valid. That is

$$1^2 + 2^2 + 3^2 + \cdots + k^2 = \frac{k(k+1)(2k+1)}{6}. \quad (1)$$

We want to prove that the statement $P(k+1)$ is also valid. That is

$$1^2 + 2^2 + 3^2 + \cdots + k^2 + (k+1)^2 = \frac{(k+1)(k+2)(2k+3)}{6}. \quad (2)$$

Using the induction hypothesis (3), the left-hand side of (4) is

$$\begin{aligned} 1^2 + 2^2 + 3^2 + \cdots + k^2 + (k+1)^2 &= \frac{k(k+1)(2k+1)}{6} + (k+1)^2 \\ &= \frac{(k+1)}{6} (k(2k+1) + 6(k+1)) \\ &= \frac{(k+1)}{6} (2k^2 + 7k + 6). \end{aligned}$$

On the other hand, by expanding the last two factors we find that the right-hand side of (4) is

$$\begin{aligned} \frac{(k+1)(k+2)(2k+3)}{6} &= \frac{(k+1)}{6} (k+2)(2k+3) \\ &= \frac{(k+1)}{6} (2k^2 + 7k + 6). \end{aligned}$$

Hence, the two sides are equal and this proves that $P(k+1)$ is valid under the assumption that $P(k)$ is valid.

Conclusion: Therefore, by the principle of mathematical induction, the statement $P(n)$ is valid for any $n \geq 1$.

2. Use induction to show that for $n \geq 1$ we have

$$1^3 + 2^3 + 3^3 + \cdots + n^3 = \frac{n^2(n+1)^2}{4}.$$

Solution: Let us denote by $P(n)$ the statement

$$1^3 + 2^3 + 3^3 + \cdots + n^3 = \frac{n^2(n+1)^2}{4}.$$

We want to show that $P(n)$ is valid for any $n \geq 1$.

Base case for $n = 1$: the left-hand side is $1^3 = 1$. The right-hand side is $\frac{1^2 \cdot (1+1)^2}{4} = 1$. Hence the equation is valid for $n = 1$.

Induction step: We suppose that for some integer $k \geq 1$, the statement $P(k)$ is valid. That is

$$1^3 + 2^3 + 3^3 + \cdots + k^3 = \frac{k^2(k+1)^2}{4}. \quad (3)$$

We want to prove that the statement $P(k+1)$ is also valid. That is

$$1^3 + 2^3 + 3^3 + \cdots + k^3 + (k+1)^3 = \frac{(k+1)^2(k+2)^2}{4}. \quad (4)$$

Using the induction hypothesis (3), the left-hand side of (4) is

$$\begin{aligned} 1^3 + 2^3 + 3^3 + \cdots + k^3 + (k+1)^3 &= \frac{k^2(k+1)^2}{4} + (k+1)^3 \\ &= \frac{(k+1)^2}{4} (k^2 + 4(k+1)) \\ &= \frac{(k+1)^2}{4} (k^2 + 4k + 4). \end{aligned}$$

On the other hand, by expanding the factor $(k+2)^2$ we find that the right-hand side of (4) is

$$\frac{(k+1)^2(k+2)^2}{4} = \frac{(k+1)^2}{4} (k^2 + 4k + 4).$$

Hence, the two sides are equal and this proves that $P(k+1)$ is valid under the assumption that $P(k)$ is valid.

Conclusion: Therefore, by the principle of mathematical induction, the statement $P(n)$ is valid for any $n \geq 1$.

3. Use induction to show that for $n \geq 1$ we have

$$1 + 3 + 6 + 10 + \cdots + \frac{n(n+1)}{2} = \frac{n(n+1)(n+2)}{6}.$$

Solution: Let us denote by $P(n)$ the statement

$$1 + 3 + 6 + 10 + \cdots + \frac{n(n+1)}{2} = \frac{n(n+1)(n+2)}{6}.$$

We want to show that $P(n)$ is valid for any $n \geq 1$.

Base case for $n = 1$: the left-hand side is 1. The right-hand side is $\frac{1 \cdot (1+1)(1+2)}{6} = 1$. Hence the equation is valid for $n = 1$.

Induction step: We suppose that for some integer $k \geq 1$, the statement $P(k)$ is valid. That is

$$1 + 3 + 6 + 10 + \cdots + \frac{k(k+1)}{2} = \frac{k(k+1)(k+2)}{6}. \quad (5)$$

We want to prove that the statement $P(k+1)$ is also valid. That is

$$1 + 3 + 6 + 10 + \cdots + \frac{k(k+1)}{2} + \frac{(k+1)(k+2)}{2} = \frac{(k+1)(k+2)(k+3)}{6}. \quad (6)$$

Using the induction hypothesis (5), the left-hand side of (6) is

$$\begin{aligned} 1 + 3 + 6 + 10 + \cdots + \frac{k(k+1)}{2} + \frac{(k+1)(k+2)}{2} &= \frac{k(k+1)(k+2)}{6} + \frac{(k+1)(k+2)}{2} \\ &= \frac{k(k+1)(k+2)}{6} + \frac{3(k+1)(k+2)}{6} \\ &= \frac{(k+3)k(k+1)(k+2)}{6}. \end{aligned}$$

This proves that $P(k+1)$ is valid under the assumption that $P(k)$ is valid.

Conclusion: Therefore, by the principle of mathematical induction, the statement $P(n)$ is valid for any $n \geq 1$.

4. Use induction to show that for $n \geq 1$ we have

$$2 + 2^3 + 2^5 + \cdots + 2^{2n-1} = \frac{2(2^{2n} - 1)}{3}.$$

Solution: Let us denote by $P(n)$ the statement

$$2 + 2^3 + 2^5 + \cdots + 2^{2n-1} = \frac{2(2^{2n} - 1)}{3}.$$

We want to show that $P(n)$ is valid for any $n \geq 1$.

Base case for $n = 1$: the left-hand side is $2^1 = 2$. The right-hand side is $\frac{2(2^{2 \cdot 1} - 1)}{3} = 2$. Hence the equation is valid for $n = 1$.

Induction step: We suppose that for some integer $k \geq 1$, the statement $P(k)$ is valid. That

is

$$2 + 2^3 + 2^5 + \dots + 2^{2k-1} = \frac{2(2^{2k} - 1)}{3}. \quad (7)$$

We want to prove that the statement $P(k+1)$ is also valid. That is

$$2 + 2^3 + 2^5 + \dots + 2^{2k+1} = \frac{2(2^{2k+2} - 1)}{3}. \quad (8)$$

Using the induction hypothesis (7), the left-hand side of (8) is

$$\begin{aligned} 2 + 2^3 + 2^5 + \dots + 2^{2k-1} + 2^{2k+1} &= \frac{2(2^{2k} - 1)}{3} + 2^{2k+1} \\ &= \frac{2^{2k+1} - 2 + 3 \cdot 2^{2k+1}}{3} \\ &= \frac{4 \cdot 2^{2k+1} - 2}{3} \\ &= \frac{2(2^{2k+2} - 1)}{3}. \end{aligned}$$

This proves that $P(k+1)$ is valid under the assumption that $P(k)$ is valid.

Conclusion: Therefore, by the principle of mathematical induction, the statement $P(n)$ is valid for any $n \geq 1$.

5. * Use induction to show that for $n \geq 1$ we have

$$(3n+1) + (3n+4) + (3n+7) + \dots + (6n-2) = \frac{n(9n-1)}{2}.$$

Solution: Let us denote by $P(n)$ the statement

$$(3n+1) + (3n+4) + (3n+7) + \dots + (6n-2) = \frac{n(9n-1)}{2}.$$

We want to show that $P(n)$ is valid for any $n \geq 1$.

Base case for $n = 1$: the left-hand side is $(3 \cdot 1 + 1) = 4$. The right-hand side is $\frac{1(9 \cdot 1 - 1)}{2} = 4$. Hence the equation is valid for $n = 1$.

Induction step: We suppose that for some integer $k \geq 1$, the statement $P(k)$ is valid. That is

$$(3k+1) + (3k+4) + (3k+7) + \dots + (6k-2) = \frac{k(9k-1)}{2}. \quad (9)$$

We want to prove that the statement $P(k+1)$ is also valid. That is

$$(3k+4) + (3k+7) + (3k+10) + \dots + (6k+4) = \frac{(k+1)(9k+8)}{2}. \quad (10)$$

Using the induction hypothesis (9), the left-hand side of (10) is

$$\begin{aligned}
 & (3k+4) + (3k+7) + (3k+10) + \cdots + (6k-2) + (6k+1) + (6k+4) \\
 &= -(3k+1) + \underbrace{\left((3k+1) + (3k+4) + (3k+7) + (3k+10) + \cdots + (6k-2) \right)}_{\text{LHS of (9)}} + (6k+1) + (6k+4) \\
 &= -(3k+1) + \frac{k(9k-1)}{2} + (6k+1) + (6k+4) \\
 &= \frac{k(9k-1)}{2} + (9k+4) \\
 &= \frac{k(9k-1) + 18k + 8}{2} \\
 &= \frac{9k^2 + 17k + 8}{2}.
 \end{aligned}$$

On the other hand, the right-hand side of (10) is

$$\frac{(k+1)(9k+8)}{2} = \frac{9k^2 + 17k + 8}{2}.$$

Hence, the two sides are equal and this proves that $P(k+1)$ is valid under the assumption that $P(k)$ is valid.

Conclusion: Therefore, by the principle of mathematical induction, the statement $P(n)$ is valid for any $n \geq 1$.

6. Use induction to show that 7 divides $8^n - 1$ for $n \geq 1$.

Solution: Let us denote by $P(n)$ the statement that 7 divides n . We want to show that $P(n)$ is valid for any $n \geq 1$.

Base case for $n = 1$: It is immediate that 7 divides $8^1 - 1 = 7$. Hence $P(1)$ is true.

Induction step: We suppose that for some integer $k \geq 1$, the statement $P(k)$ is valid. That is 7 divides $8^k - 1$. We want to prove that the statement $P(k+1)$ is also valid. That is 7 divides $8^{k+1} - 1$. We write

$$\begin{aligned}
 8^{k+1} - 1 &= 8 \cdot 8^k - 1 \\
 &= 8 \cdot (8^k - 1) + 8 - 1 \\
 &= 8 \cdot (8^k - 1) + 7.
 \end{aligned}$$

By our assumption, we know that 7 divides $8^k - 1$. Hence 7 divides $8 \cdot (8^k - 1) + 7$. This proves that $P(k+1)$ is valid under the assumption that $P(k)$ is valid.

Conclusion: Therefore, by the principle of mathematical induction, the statement $P(n)$ is valid for any $n \geq 1$.

7. * Find the fallacy in the following inductive argument to prove that "all birds are the same colour":

Base case when $n = 1$: In a set of birds consisting of only one bird, all birds are clearly the same colour. Hence the result is valid when the number of birds in a set is $n = 1$.

Induction step: Suppose now that the result is valid when a set of birds contains k birds; that is all k birds are the same colour. Consider any set of $k + 1$ birds, and enumerate the birds with labels $1, 2, \dots, k + 1$. The set of birds with labels $1, 2, \dots, k$ must all be the same colour by the inductive hypothesis. The set of birds $2, 3, \dots, k + 1$ must also all be the same colour. Since these two sets have $k - 1$ birds in common (namely the one with labels $2, 3, \dots, k - 1$), all $k + 1$ birds must have the same colour.

Conclusion: Thus, by the principle of mathematical induction, any set of n birds must all be the same colour.

Solution: The base case is correct. The induction step is also correct, but only when $k \geq 2$. This is because the intersection of the two sets has size $k - 1$, which is 0 when $k = 1$. So the problematic step is the step from $k = 1$ to $k = 2$. The statement for $k = 2$ is not true, because we can have two birds with different colours. In order to apply the principle mathematical induction, the induction step has to be true for any $k \geq 1$, and not only $k \geq 2$.

8. * Use induction to show that for $n \geq 1$ we have

$$1 + 3 + 5 + \dots + (2n - 1) < n^2 + n.$$

Solution: Let us denote by $P(n)$ the statement

$$1 + 3 + 5 + \dots + (2n - 1) < n^2 + n.$$

We want to show that $P(n)$ is valid for any $n \geq 1$.

Base case for $n = 1$: the left-hand side is 1. The right-hand side of the inequality is $1^2 + 1 = 2$. Since $1 < 2$ is true, $P(1)$ is true.

Induction step: We suppose that for some integer $k \geq 1$, the statement $P(k)$ is valid. That is

$$1 + 3 + 5 + \dots + (2k - 1) < k^2 + k. \quad (11)$$

We want to prove that the statement $P(k + 1)$ is also valid. That is

$$1 + 3 + 5 + \dots + (2k + 1) < (k + 1)^2 + (k + 1) = k^2 + 3k + 2. \quad (12)$$

Using the induction hypothesis (11), the left-hand side of (12) is

$$\begin{aligned} 1 + 3 + 5 + \dots + (2k - 1) + (2k + 1) &< k^2 + k + (2k + 1) \\ &= k^2 + 3k + 1 \\ &< k^2 + 3k + 2. \end{aligned}$$

This proves that $P(k + 1)$ is valid under the assumption that $P(k)$ is valid.

Conclusion: Therefore, by the principle of mathematical induction, the statement $P(n)$ is valid for any $n \geq 1$.

9. * Use induction to show that $9(n!) > 2^{2n}$ for $n \geq 5$.

Solution: Let us denote by $P(n)$ the statement

$$9(n!) > 2^{2n}.$$

We want to show that $P(n)$ is valid for any $n \geq 5$.

Base case for $n = 5$: the left-hand side is $9 \cdot 5! = 9 \cdot 120 = 1080$. The right-hand side of the inequality is $2^{2 \cdot 5} = 2^{10} = 1024$. Since $1080 > 1024$ is true, $P(5)$ is true.

Induction step: We suppose that for some integer $k \geq 1$, the statement $P(k)$ is valid. That is

$$9(k!) > 2^{2k}. \quad (13)$$

We want to prove that the statement $P(k+1)$ is also valid. That is

$$9((k+1)!) > 2^{2k+2}. \quad (14)$$

Using the induction hypothesis (13), the left-hand side of (14) is

$$\begin{aligned} 9((k+1)!) &= 9(k+1) \cdot (k!) \\ &\geq 6 \cdot 2^{2k} \\ &> 4 \cdot 2^{2k} = 2^{2k+2}. \end{aligned}$$

We used that $k \geq 5$, so $k+1 \geq 6$. This proves that $P(k+1)$ is valid under the assumption that $P(k)$ is valid.

Conclusion: Therefore, by the principle of mathematical induction, the statement $P(n)$ is valid for any $n \geq 1$.

10. Express the sum in sigma notation

$$\frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{4}{16} + \frac{5}{32} + \cdots + \frac{10}{1024}.$$

Initiate the summation with the integer 1.

Solution: We see that as we go through the sum from left to right, the numerator increases by 1 and the denominator is multiplied by 2. So the terms that we are summing are of the shape $\frac{k}{2^k}$ and k is between 1 and 10. We can check that this formula is actually correct: for $k = 1$ we obtain $\frac{1}{2^1} = \frac{1}{2}$, for $k = 2$ we obtain $\frac{2}{2^2} = \frac{2}{4}$, for $k = 3$ we obtain $\frac{3}{2^3} = \frac{3}{8}$, ..., for $k = 10$ we obtain $\frac{10}{2^{10}} = \frac{10}{1024}$. Hence, the sigma notation initiated with 1 is

$$\frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{4}{16} + \frac{5}{32} + \cdots + \frac{10}{1024} = \sum_{k=1}^{10} \frac{k}{2^k}.$$

11. Express the sums that appear in questions 1,2, 3,4,5 in sigma notation.

Solution: Possible sigma notations are

1. $\sum_{k=1}^n k^2,$
2. $\sum_{k=1}^n k^3,$
3. $\sum_{k=1}^n \frac{k(k+1)}{2},$
4. $\sum_{k=1}^n 2^{2k-1},$
5. $\sum_{k=n}^{2n-1} (3k+1).$

12. Express the sum in sigma notation

$$2 \cdot 3 + 3 \cdot 4 + 4 \cdot 5 + \cdots + 99 \cdot 100.$$

Initiate the summation with the integer 1.

Solution: The sigma notation starting at 1 is

$$\sum_{k=1}^{98} (k+1)(k+2).$$

13. Verify that the two summations are identical

$$\sum_{j=1}^{14} 3^{j+5}(j+4)! \quad \text{and} \quad \sum_{j=5}^{18} 3^{j+1}j!$$

Solution: If we do the substitution $m = j + 4$ then the sum on the left is

$$\sum_{j=1}^{14} 3^{j+5}(j+4)! = \sum_{m=5}^{18} 3^{m+1}m!$$

By replacing the index of notation m by j we have

$$\sum_{m=5}^{18} 3^{m+1}m! = \sum_{j=5}^{18} 3^{j+1}j!$$

14. Use the formulas of questions 3 and 4 to evaluate the sum

$$\sum_{k=1}^5 \frac{4^k + k(k+1)}{2}.$$

Solution: We write

$$\begin{aligned}
 \sum_{k=1}^5 \frac{4^k + k(k+1)}{2} &= \sum_{k=1}^5 \frac{4^k}{2} + \sum_{k=1}^5 \frac{k(k+1)}{2} \\
 &= \sum_{k=1}^5 2^{2k-1} + \sum_{k=1}^5 \frac{k(k+1)}{2} \\
 &= \frac{2(2^{10} - 1)}{3} + \frac{5(5+1)(5+2)}{6} \\
 &= 682 + 35 \\
 &= 717.
 \end{aligned}$$

15. Use the formulas of questions 1 and 2 to evaluate the sum

$$\sum_{k=2}^{29} (k^3 - 3k^2).$$

Solution: We know that

$$\begin{aligned}
 \sum_{k=2}^{29} k^3 &= 2^3 + 3^3 + \cdots + 29^3 \\
 &= -1 + (1^3 + 2^3 + 3^3 + \cdots + 29^3) \\
 &= -1 + \frac{29^2 \cdot 30^2}{4}.
 \end{aligned}$$

We also know that

$$\begin{aligned}
 \sum_{k=2}^{29} k^2 &= 2^2 + 3^2 + \cdots + 29^2 \\
 &= -1 + (1^2 + 2^2 + 3^2 + \cdots + 29^2) \\
 &= -1 + \frac{29 \cdot 30 \cdot 59}{6}.
 \end{aligned}$$

Hence

$$\begin{aligned}
 \sum_{k=2}^{29} (k^3 - 3k^2) &= \sum_{k=2}^{29} k^3 - 3 \sum_{k=2}^{29} k^2 \\
 &= -1 + \frac{29^2 \cdot 30^2}{4} - 3 \left(-1 + \frac{29 \cdot 30 \cdot 59}{6} \right) \\
 &= 2 + 29^2 \cdot 15^2 - 29 \cdot 15 \cdot 59 \\
 &= 163562.
 \end{aligned}$$

16. * Find a formula for the summation

$$\sum_{k=1}^n \frac{1}{k(k+1)}.$$

Hint: $\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}$.

Solution: Using the hint we write

$$\begin{aligned}\sum_{k=1}^n \frac{1}{k(k+1)} &= \sum_{k=1}^n \left(\frac{1}{k} - \frac{1}{k+1} \right) \\&= \sum_{k=1}^n \frac{1}{k} - \sum_{k=1}^n \frac{1}{k+1} \\&= \sum_{k=1}^n \frac{1}{k} - \sum_{k=2}^{n+1} \frac{1}{k} \\&= 1 + \sum_{k=2}^n \frac{1}{k} - \left(\sum_{k=2}^n \frac{1}{k} + \frac{1}{n+1} \right) \\&= 1 - \frac{1}{n+1} \\&= \frac{n}{n+1}.\end{aligned}$$